

Chapter 12.

§12.1.

↙ Cartesian/rectangular coordinates

Distance between $P_1(x_1, y_1, z_1)$ and $P_2(x_2, y_2, z_2)$ is

$$|P_1 P_2| = \sqrt{(x_1 - x_2)^2 + (y_1 - y_2)^2 + (z_1 - z_2)^2}$$

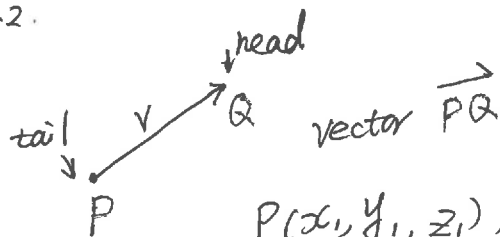
Standard equation for the sphere in space

Center (x_0, y_0, z_0) , radius a .

$$(x - x_0)^2 + (y - y_0)^2 + (z - z_0)^2 = a^2$$

§12.2.

Two vectors are equal if they have the same length and direction.



$P(x_1, y_1, z_1), Q(x_2, y_2, z_2)$

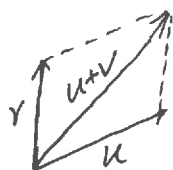
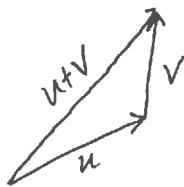
$$V = \vec{PQ} = \langle \underbrace{x_2 - x_1}_{V_1}, \underbrace{y_2 - y_1}_{V_2}, \underbrace{z_2 - z_1}_{V_3} \rangle$$

Vector Magnitude or length: $|V| = |\vec{PQ}| = \sqrt{V_1^2 + V_2^2 + V_3^2}$

$u = \langle u_1, u_2, u_3 \rangle, V = \langle V_1, V_2, V_3 \rangle, K$ a scalar

Vector addition: $u + V = \langle u_1 + V_1, u_2 + V_2, u_3 + V_3 \rangle$

Scalar multiplication: $Ku = \langle Ku_1, Ku_2, Ku_3 \rangle$



— parallelogram law

Properties of vector operations

u, v, w vectors, a, b scalars

1) $u + v = v + u$ 2) $(u + v) + w = u + (v + w)$ 3) $u + 0 = u$

4) $u + (-u) = 0$ 5) $0u = 0$ 6) $|u| = u$

7. $a(bu) = (ab)u$ 8) $a(u + v) = au + av$ 9) $(a + b)u = au + bu$

unit vector: $|u| = 1$

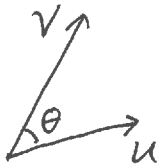
For any nonzero vector V , $\boxed{u = \frac{V}{|V|}}$ is a unit vector in the direction of V .

§12.3.

Given two vectors $u = \langle u_1, u_2, u_3 \rangle$, $v = \langle v_1, v_2, v_3 \rangle$, their dot product is

$$u \cdot v = u_1 v_1 + u_2 v_2 + u_3 v_3$$

Clearly, $|u|^2 = u \cdot u$



The angle between two nonzero vector u and v

$$\theta = \cos^{-1}\left(\frac{u \cdot v}{|u||v|}\right) \quad \leftarrow \text{the cosine formula.}$$

$$0 \leq \theta \leq \pi$$

$$\Rightarrow \begin{cases} \text{i) } u \cdot v > 0 \Leftrightarrow 0 \leq \theta < \frac{\pi}{2} \\ \text{ii) } u \cdot v < 0 \Leftrightarrow \frac{\pi}{2} < \theta \leq \pi \\ \text{iii) } u \cdot v = 0 \Leftrightarrow \theta = \frac{\pi}{2} \end{cases}$$

Properties of dot product

1) $u \cdot v = v \cdot u$

2) $(\alpha u) \cdot v = u \cdot (\alpha v) = \alpha(u \cdot v)$, α is a scalar

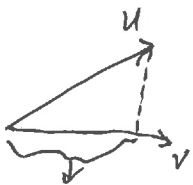
3) $(u+v) \cdot w = u \cdot w + v \cdot w$

4) $0 \cdot u = 0$

Orthogonal vectors

$$u \perp v \Leftrightarrow u \cdot v = 0 \quad (\text{i.e., } \theta = \frac{\pi}{2})$$

Vector projection

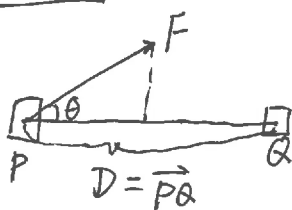


$$\text{Proj}_v u = \left(\frac{u \cdot v}{|v|^2}\right) v$$

scalar component of u in the direction of v : $|u| \cos \theta = \frac{u \cdot v}{|v|}$

$\text{Proj}_v u$ - vector projection of u onto v

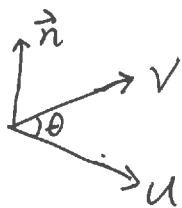
Work



$$\text{Work} = (\text{Scalar component of } u \text{ in the direction of } D) \cdot (\text{Length of } D)$$

$$= F \cdot D$$

§12.4



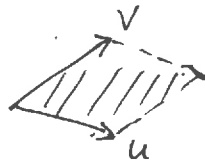
The cross product

$$u \times v = (|u||v|\sin\theta) \hat{n}$$

unit vector perpendicular to the plane of u & v with the right-hand rule.

{ i) Nonzero vectors u and v are parallel $\Leftrightarrow u \times v = 0$

{ ii) $|u \times v|$ is the area of the parallelogram determined by u & v



properties of cross product

u, v, w vectors, r, s scalars

1) $(ru) \times (sv) = (rs)(u \times v)$

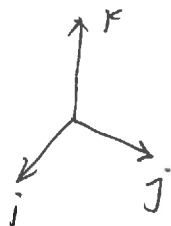
2) $u \times (v + w) = u \times v + u \times w$

3) $v \times u = -(u \times v)$

4) $(v + w) \times u = v \times u + w \times u$

5) $0 \times u = 0$

6) $u \times (v \times w) = (u \cdot w)v - (u \cdot v)w$



$i \times j = k = -(j \times i)$

$j \times k = i = -(k \times j)$

$k \times i = j = -(i \times k)$

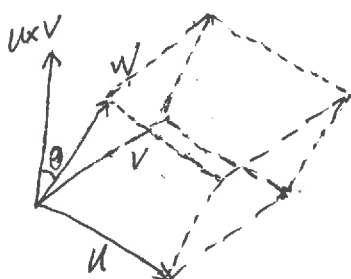
calculating formula

$$u \times v = \begin{vmatrix} i & j & k \\ u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \end{vmatrix} = i \begin{vmatrix} u_2 & u_3 \\ v_2 & v_3 \end{vmatrix} - j \begin{vmatrix} u_1 & u_3 \\ v_1 & v_3 \end{vmatrix} + k \begin{vmatrix} u_1 & u_2 \\ v_1 & v_2 \end{vmatrix}$$

$$= i(u_2v_3 - u_3v_2) - j(u_1v_3 - u_3v_1) + k(u_1u_2 - u_2v_1)$$

Triple scalar product

$$(u \times v) \cdot w = |u \times v| |w| \cos\theta$$



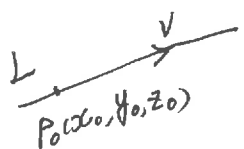
$|(u \times v) \cdot w| = \text{Volume of the parallelepiped.}$

calculating formula

$$(u \times v) \cdot w = \begin{vmatrix} u_1 & u_2 & u_3 \\ v_1 & v_2 & v_3 \\ w_1 & w_2 & w_3 \end{vmatrix}$$

§12.5.

Vector equation for a line L through $P_0(x_0, y_0, z_0)$ parallel to V is

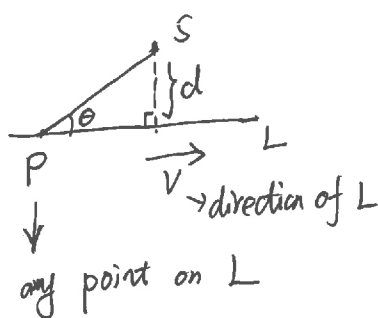


$$\begin{aligned}
 Y(t) &= \overrightarrow{OP_0} + tV \\
 &= r_0 + tV = \langle x_0 + tV_1, y_0 + tV_2, z_0 + tV_3 \rangle
 \end{aligned}$$

Parametric equation of the Line L

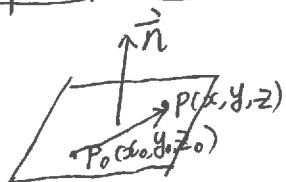
$$x = x_0 + tV_1, \quad y = y_0 + tV_2, \quad z = z_0 + tV_3, \quad -\infty < t < \infty$$

Distance from a point S to a line L



$$\begin{aligned}
 d &= |\overrightarrow{PS}| \sin \theta \\
 &= \frac{|\overrightarrow{PS} \times V|}{|V|}
 \end{aligned}$$

Equation for a plane in Space

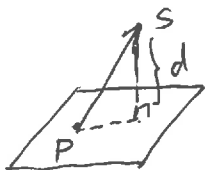


The plane through $P_0(x_0, y_0, z_0)$ normal to $\vec{n} = A\mathbf{i} + B\mathbf{j} + C\mathbf{k}$.

$$\vec{n} \cdot \overrightarrow{P_0P} = 0$$

$$A(x - x_0) + B(y - y_0) + C(z - z_0) = 0$$

Distance from a point S to a plane



$$d = \left| \overrightarrow{PS} \cdot \frac{\vec{n}}{|\vec{n}|} \right| \quad \text{where } \vec{n} \text{ is a normal vector of the plane.}$$

Angle between two plane.

$$\theta = \cos^{-1} \left(\frac{|\vec{n}_1 \cdot \vec{n}_2|}{|\vec{n}_1| |\vec{n}_2|} \right)$$

where $\vec{n}_1, \vec{n}_2$ are normal vectors of the two planes respectively.

§ 12.6.

Quadric surface : the graph of a second degree equation in x, y, z

$$Ax^2 + By^2 + Cz^2 + Dxy + Eyz + Fxz + Gx + Hy + Iz + J = 0$$

↳ General form

⇓ removing translations and rotations

$$\underline{Ax^2 + By^2 + Cz^2 + Dz = E}$$

Ellipsoid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$ ($a=b=c \rightarrow$ sphere)

Elliptical paraboloid: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z}{c}$

Elliptical Cone: $\frac{x^2}{a^2} + \frac{y^2}{b^2} = \frac{z^2}{c^2}$

Hyperboloid of one sheet: $\frac{x^2}{a^2} + \frac{y^2}{b^2} - \frac{z^2}{c^2} = 1$

Hyperboloid of two sheets: $-\frac{x^2}{a^2} - \frac{y^2}{b^2} + \frac{z^2}{c^2} = 1$

Hyperbolic paraboloid: $\frac{y^2}{b^2} - \frac{x^2}{a^2} = \frac{z}{c}, \quad c > 0$

Chapter 13.

§13.1.

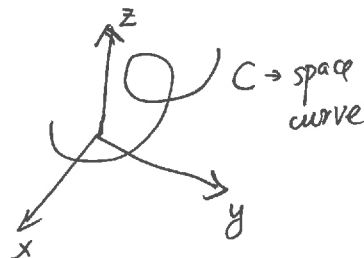
Vector-valued function: a rule that assigns a vector in space to each element in the domain D .

space curves $\rightarrow D$: the intervals of real numbers

Vector equation: $\vec{r}(t) = \langle f(t), g(t), h(t) \rangle, t \in I$

$$\downarrow$$

$$x = f(t), y = g(t), z = h(t)$$



1. $\lim_{t \rightarrow t_0} \vec{r}(t) = \langle \lim_{t \rightarrow t_0} f(t), \lim_{t \rightarrow t_0} g(t), \lim_{t \rightarrow t_0} h(t) \rangle$ if all of them exist.

2. $\vec{r}(t)$ is continuous at t_0 if $\lim_{t \rightarrow t_0} \vec{r}(t) = \vec{r}(t_0)$

3. $\vec{r}'(t) = \frac{d\vec{r}}{dt} = \langle f'(t), g'(t), h'(t) \rangle$ ← derivative of $\vec{r}$ at t .

Differential rules:

1. $\frac{d}{dt} C = 0$

2. $\frac{d}{dt} [\alpha u(t)] = \alpha u'(t), \frac{d}{dt} [f(t)u(t)] = f'(t)u(t) + f(t)u'(t)$

3. $\frac{d}{dt} [u(t) + v(t)] = u'(t) + v'(t)$

4. $\frac{d}{dt} [u(t) \cdot v(t)] = u'(t) \cdot v(t) + u(t) \cdot v'(t)$

5. $\frac{d}{dt} [u(t) \times v(t)] = u'(t) \times v(t) + u(t) \times v'(t)$

6. $\frac{d}{dt} [u(f(t))] = f'(t)u'(f(t))$

Remark: If $|\vec{r}(t)| = \text{Constant}$, then $\vec{r}(t) \perp \vec{r}'(t)$.

position vector of a particle in space

$\vec{r}(t) \Rightarrow$ velocity: $\vec{v} = \frac{d\vec{r}}{dt}$

speed: $|\vec{v}| \rightarrow \text{direction of motion } \frac{\vec{v}}{|\vec{v}|}$

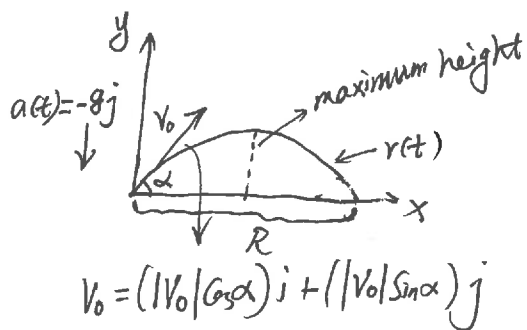
Acceleration: $\vec{a} = \frac{d\vec{v}}{dt} = \frac{d^2\vec{r}}{dt^2}$

§13.2.

$$\begin{aligned}\int r(t) dt &= \left(\int f(t) dt \right) i + \left(\int g(t) dt \right) j + \left(\int h(t) dt \right) k \\ &= \underbrace{F(t) i + G(t) j + H(t) k}_{R(t)} + C\end{aligned}$$

$$\begin{aligned}\int_a^b r(t) dt &= \left(\int_a^b f(t) dt \right) i + \left(\int_a^b g(t) dt \right) j + \left(\int_a^b h(t) dt \right) k \\ &= [F(b) - F(a)] i + [G(b) - G(a)] j + [H(b) - H(a)] k \\ &= R(b) - R(a)\end{aligned}$$

Projectile motion



initial position $r_0 = 0$
initial velocity v_0

Idea projectile Motion

$$\begin{aligned}r(t) &= -\frac{1}{2} g t^2 j + (|v_0| \cos \alpha i + |v_0| \sin \alpha j) t \\ &= (|v_0| t \cos \alpha) i + (|v_0| t \sin \alpha - \frac{1}{2} g t^2) j\end{aligned}$$

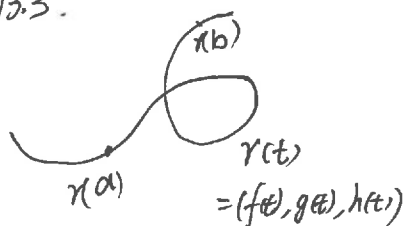
Time to reach the maximum height $y_{\max}$

$$t = \frac{|v_0| \sin \alpha}{g}$$

maximum height: $y_{\max} = \frac{(|v_0| \sin \alpha)^2}{2g}$

range: $R = \frac{|v_0|^2 \sin 2\alpha}{g}$

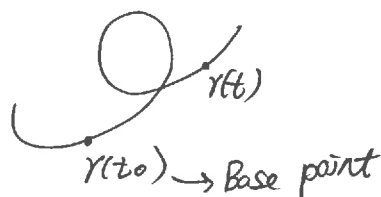
§13.3.



length of the arc from $r(a)$ to $r(b)$

$$\begin{aligned}L &= \int_a^b |r'(t)| dt \\ &= \int_a^b \sqrt{(f'(t))^2 + (g'(t))^2 + (h'(t))^2} dt\end{aligned}$$

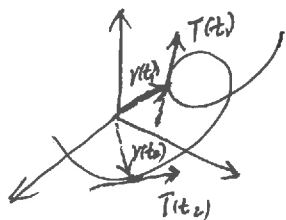
Arc length function: $s(t) = \int_0^t |r'(u)| du$
 $\Rightarrow \frac{ds}{dt} = |r'(t)|$



Unit tangent vector

$$T(t) = \frac{r'(t)}{|r'(t)|} = \frac{v(t)}{|v(t)|}$$

→ physical meaning: $T = \frac{dr}{ds} \rightarrow$ rate of change in r with respect to the arc length.



§13.4

Curvature of the curve

$$k = \left| \frac{dT}{ds} \right| \rightarrow \text{physical meaning: } k = \left| \frac{d^2r}{ds^2} \right|$$

calculating formula: $k = \frac{|T'(t)|}{|r'(t)|}$

Principal unit normal vector

$$N = \frac{1}{k} \frac{dT}{ds} \rightarrow \text{physical meaning: } \frac{dT/ds}{|dT/ds|}$$

calculating formula: $N = \frac{T'(t)}{|T'(t)|}$

$$\boxed{T \perp N}$$

→ Since T is a vector with constant length, $|T| = 1$.



Chapter 14.

§ 14.1.

A real-valued function f on D

$$W = f(x_1, x_2, \dots, x_n) \rightarrow \text{n-tuples of real numbers}$$

Domain: D

$$\text{Range: } \{f(x_1, x_2, \dots, x_n) \mid (x_1, x_2, \dots, x_n) \in D\}$$

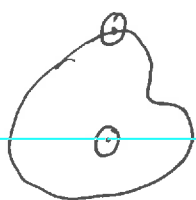
Given a region R , (x_0, y_0) is

- interior point: if it the center of a disk of positive radius that lies entirely in R
- boundary point: if every disk centered at (x_0, y_0) contains points that lies outside of R as well as points that lies in R .

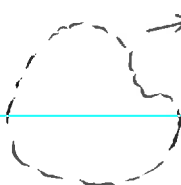
R is

- open: if it consists entirely of interior points
- closed: if it contains all its boundary points.

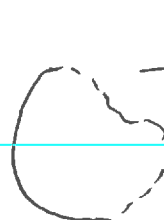
bounded
unbounded



← closed



→ open



→ neither closed nor open

Remark: Extension to 3D space for (x_0, y_0, z_0)

$(x, y, f(x, y)) \leftarrow$ the graph of the function f or the surface $z = f(x, y)$

- "Level curve" $f(x, y) = C \leftarrow$ function of two variable
- "Level surface" $f(x, y, z) = C \leftarrow$ function of three variables.

§ 14.2.

$$\lim_{(x, y) \rightarrow (x_0, y_0)} f(x, y) = L$$

$\Leftrightarrow$

for every number $\varepsilon > 0$, there exists a corresponding number $\delta > 0$ such that for all (x, y) in the domain of f
 $|f(x, y) - L| < \varepsilon$ whenever $0 < \sqrt{(x - x_0)^2 + (y - y_0)^2} < \delta$.

Similarly, $\lim_{(x, y, z) \rightarrow (x_0, y_0, z_0)} f(x, y, z) = L$ is defined!

Properties: If $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) = L$ and $\lim_{(x,y) \rightarrow (x_0,y_0)} g(x,y) = M$

Then 1. $\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y) \pm g(x,y)) = L \pm M$

2. $\lim_{(x,y) \rightarrow (x_0,y_0)} k f(x,y) = k L$ (any number k)

3. $\lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) g(x,y) = L M$

4. $\lim_{(x,y) \rightarrow (x_0,y_0)} \frac{f(x,y)}{g(x,y)} = \frac{L}{M}$

5. $\lim_{(x,y) \rightarrow (x_0,y_0)} (f(x,y))^n = L^n$, n a positive integer.

6. $\lim_{(x,y) \rightarrow (x_0,y_0)} \sqrt[n]{f(x,y)} = \sqrt[n]{L}$, n a positive integer; if n is even, assume $L > 0$.

Remark: For $\frac{0}{0}$ type limits, try $y = kx$ or $y = kx^2$ types of tests with different values of k .

$f(x,y)$ is continuous at (x_0, y_0)

$$\Leftrightarrow \begin{cases} f \text{ is defined at } (x_0, y_0) \\ \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y) \text{ exists} \\ f(x_0, y_0) = \lim_{(x,y) \rightarrow (x_0,y_0)} f(x,y). \end{cases}$$

★ $f(x,y)$ is a continuous function if it is continuous at every point of its domain.

Continuity of Composites

f is continuous at (x_0, y_0) , g is continuous at $f(x_0, y_0)$

$$\Rightarrow h = g \circ f = g(f(x,y)) \text{ is continuous at } (x_0, y_0)$$